

Paper Practice 1.4 • WORKED KEY**The Distributive Property**

Show all of your work in the space provided. Write each answer in simplest form. Circle your final answers.

1. Use the distributive property to rewrite and evaluate each expression using mental math. (Mental Math)

a. $7(43)$

Answer: $7(40 + 3) = 280 + 21 = 301$

b. $9(196)$

Answer: $9(200 - 4) = 1800 - 36 = 1764$

c. $6(5 \frac{1}{6})$

Answer: $6(5 + \frac{1}{6}) = 30 + 1 = 31$

2. Use the distributive property to simplify each expression. (Distribute)

a. $3(4x - 9)$

Answer: $12x - 27$

b. $(2y^2 + 5y - 3)4$

Answer: $8y^2 + 20y - 12$

c. $-5(2a - 3b + 1)$

Answer: $-10a + 15b - 5$ – the negative hits EVERY term

3. Simplify, if possible: $15n - 4n$ (Like Terms)

Answer: $11n$

4. Simplify, if possible: $5c^2 + 7c - 2c$ (Like Terms)

Answer: $5c^2 + 5c - c^2$ and c are NOT like terms

5. Simplify, if possible: $x^2 + 9x + 9$ (Like Terms)

Answer: Already in simplest form – there are no like terms to combine.

6. Simplify: $8d + 3(2d + 7)$ (Simplify)

Answer: $8d + 6d + 21 = 14d + 21$

7. Simplify: $4(3x + 2y) - 3(5 - y)$ (Simplify)

Answer: $12x + 8y - 15 + 3y = 12x + 11y - 15$

8. Simplify: $2(3m + 7) - (m + 10)$ (Simplify)

Answer: $6m + 14 - m - 10 = 5m + 4$. The bare minus sign means -1 – distribute it to BOTH terms.

9. Write an algebraic expression for “four times the sum of $3x$ and $2y$, decreased by twice the difference of 6 and y ,” then simplify it. (Write & Simplify)

Answer: $4(3x + 2y) - 2(6 - y) = 12x + 8y - 12 + 2y = 12x + 10y - 12$

10. Write an algebraic expression for “the product of 7 and t squared, increased by 2 times the sum of 3 and t squared,” then simplify it. (Write & Simplify)

Answer: $7t^2 + 2(3 + t^2) = 7t^2 + 6 + 2t^2 = 9t^2 + 6$

11. Error Analysis – Gus simplified $-3(2x - 5)$ and wrote $-6x - 15$. Identify his mistake, then simplify correctly. (Error Analysis)

Answer: He distributed -3 to the $2x$ but kept the sign on the 5 . $(-3)(-5) = +15$. Correct: $-6x + 15$.

12. A store sells notebooks for \$3 and pens for \$2. Write an expression for the cost of n notebooks and p pens for each of 4 students, then simplify and evaluate it for $n = 3$ and $p = 5$. (Apply)

Answer: $4(3n + 2p) = 12n + 8p$; $12(3) + 8(5) = 36 + 40 = \76